

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-0934

Hood Mesa 138 kV
March 03, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0934. The Interconnection Customer proposes a 147.7 MW battery energy storage facility interconnecting at the Hood Mesa 138 kV substation in the Virginia area. The request was received on January 30, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 256 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,923,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Davis-Hood Mesa 138.0 kV	Igo 345/138 kV Autotransformer	141.3	241	88.4
Davis-Victory Way 138.0 kV	Igo 345 kV/138 kV TX	125.8	253	65.0
Mecklenburg Power Sta-Kerr Dam 345.0 kV	Bear-Mecklenburg Power Sta 345.0 kV	108.1	1,198	20.5
Thalia-Igo 138.0 kV #1	Thalia-Igo Substation 138.0 kV Ckt 2	102.1	333	24.3
West Waynesboro-Chabersburg 345.0 kV line	Urban-American Substation 345.0 kV	101.0	1,750	5.1
Yarnen Substation 345-138 kV XFMR	Mccartney-Yarnen Substation 345.0 kV	100.5	321	12.5
Richland Substation 500/230kV Autotransformer	Richland-Troy Energy Llc 500.0 kV L	100.5	2,672	6.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Davis-Hood Mesa 138.0 kV	Rebuild line with higher-capacity conductor	\$4,799,000	95% (\$4,579,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Davis-Victory Way 138.0 kV	Rebuild line with higher-capacity conductor	\$1,664,000	88% (\$1,456,000)
Mecklenburg Power Station-Kerr Dam 345.0 kV line	Reconductor line	\$12,769,000	10% (\$1,251,000)
Thalia-Igo 138.0 kV #1	Reconductor line	\$2,143,000	51% (\$1,099,000)
West Waynesboro-Chabersburg 345.0 kV line	Reconductor line	\$8,541,000	8% (\$671,000)
Yarnen Substation 345-138 kV XFMR	Replace transformer with higher-rated unit	\$20,795,000	14% (\$3,006,000)
Richland 500 kV/230 kV XFMR	Replace transformer with higher-rated unit	\$27,626,000	3% (\$861,000)
	Total allocated to project		\$12,923,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.